

The Möbius Connection as Canonical Generator of Arithmetic Structure

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Abstract

We identify a structural deficiency common to the standard constructions of the integers $\mathbb{Z}$, the reals $\mathbb{R}$, and the complex numbers $\mathbb{C}$: at each level, the axiom of choice (AC) is invoked to supply orientation information that the underlying construction does not canonically generate. The most critical instance is the Steinitz theorem, which establishes that the orientation of the complex plane is not canonically determined in the standard framework—the algebraic construction $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$ produces two indistinguishable roots of $x^2 + 1 = 0$, and no algebraic argument selects which is i .

We show that a single geometric object—the connection form $\omega = \frac{1}{2}d\theta$ on the circle S^1 , hereafter called the *Möbius connection*—provides orientation structure canonically at every level, eliminating the compensatory uses of AC and replacing arbitrary selection with structural derivation. The uniqueness of the connection is established by three intrinsic properties of flat connections on S^1 : discreteness (finite holonomy), irreducibility (prime-order holonomy group), and involutivity (holonomy squares to identity, matching the binary character of orientation). These three properties are not axioms; they are standard mathematical attributes that partition the family of flat connections into classes, with $\omega = \frac{1}{2}d\theta$ as the unique element satisfying all three simultaneously. The uniqueness is elevated to a universal property: (S^1, ω) is the initial object in the category **OAS** of oriented arithmetic structures, whose axioms formalize the already-established selection.

The connection’s canonicity is further evidenced by its extension coherence: the same generative mechanism—holonomy-driven doubling with enforced involution—extends coherently through $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$, reproducing precisely the known hierarchy of normed division algebras and their failure modes at dimensions 1, 2, 4, and 8. No competing construction is known to survive this extension constraint.

The canonical $\mathbb{C}$ generated by the connection differs from the standard algebraic $\mathbb{C}$ in carrying explicit geometric structure: two orthogonal tracks (additive and multiplicative) whose completions determine the real axis and the critical line as canonically distinguished loci. This geometric specification makes the characteristic class hierarchy of algebraic topology (Stiefel–Whitney, Chern, Pontryagin classes) interpretable as level-by-level management of ambiguities that the canonical connection resolves globally, and makes the critical line’s canonical status available as a geometric datum rather than a conventional choice.

We show that the integers arise canonically as the loop space $\pi_1(S^1)$ equipped with orientation derived from the connection; that the arithmetic Möbius function $\mu(n) = (-1)^k$ is the holonomy trace of the connection acting on prime factorizations; and that $1/\zeta(s) = \sum \mu(n)/n^s$ is a statement about holonomy eigenvalues of the multiplicative structure of $\mathbb{N}$. A companion paper applies this canonical foundation to the Riemann Hypothesis.

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1 Introduction

The complex plane $\mathbb{C}$ is the ambient space of an enormous proportion of modern mathematics. It is typically introduced as the algebraic closure of $\mathbb{R}$, or as $\mathbb{R}[i]/(i^2 + 1)$, or axiomatically as a complete algebraically closed field of characteristic zero. In all of these presentations, a crucial question goes unasked: *from what more primitive object does $\mathbb{C}$ emerge, and is that emergence canonical?*

The standard construction $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$ is algebraically complete. Every polynomial over $\mathbb{C}$ has a root. The field structure is well-defined in ZF without AC. Nothing in this construction is wrong. But the algebraic construction is *geometrically silent*: it produces the field without specifying its internal geometric constitution — how paths through the space are organized, what holonomy classes exist, how the field is generated rather than merely asserted. The construction stops at the algebraic object and leaves its geometry uncharacterized.

The first visible sign of this silence is orientation: the quotient $\mathbb{R}[x]/(x^2 + 1)$ produces two roots of $x^2 + 1 = 0$, and no algebraic argument distinguishes them. Selecting which root is called i — determining which direction is “positive imaginary” — requires a geometric datum that the algebraic construction does not provide. This is not a flaw requiring repair; it is an incompleteness requiring completion.

This paper identifies the geometric object that completes the picture: the connection form $\omega = \frac{1}{2}d\theta$ on S^1 , equipped with its $\mathbb{Z}_2$ holonomy. We call this the *Möbius connection*. It does not repair the algebraic construction — it fills its geometric silence by specifying the internal structure that the construction carries implicitly but does not express. We show that it is the unique flat connection on S^1 whose holonomy is simultaneously discrete, irreducible, and involutive; that it is the initial object in a natural category whose axioms formalize this selection; that it extends coherently through the full normed division algebra tower $\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$; and that it generates the arithmetic Möbius function as a necessary consequence.

The paper is self-contained. Its contributions stand independently as foundational results about the canonical generation of arithmetic structure.

Organization

Section 2 audits the standard constructions of $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$ for AC dependence. Section 3 introduces the Möbius connection, establishes its uniqueness by intrinsic connection-theoretic properties, formalizes the uniqueness via the category **OAS** and universal property, develops the successor-vs-loop contrast, and makes explicit the consequences of canonical $\mathbb{C}$ for complex analysis and the characteristic class hierarchy. Section 4 shows how the connection eliminates each AC invocation identified in Section 2. Section 5 develops the integers as loop space and connects to the Möbius function. Section 6 develops the reappearance of $\mathbb{C}$ within the tower and its implications. Section 7 states open problems.

Main result

The central result of this paper, established in Section 3, is the following:

Theorem (Möbius connection as canonical connection, informal). *Among all flat connections on S^1 , the Möbius connection $\omega = \frac{1}{2}d\theta$ is the unique connection whose holonomy is discrete, irreducible, and involutive. These three properties — finiteness, primality, and compatibility with the binary character of orientation — are not axioms but intrinsic attributes of flat connections. The connection is the initial object in the*

category **OAS** of oriented arithmetic structures, which formalizes the selection: every oriented arithmetic structure admits a unique morphism from (S^1, ω) .

The formal statement and proof appear as Theorem 3.9. Sections 2 and 4 establish the AC audit that motivates the question; Sections 5 and 6 develop the downstream consequences.

2 What the Standard Construction Leaves Unspecified

2.1 The Axiom of Choice as Geometric Diagnostic

The axiom of choice (AC) states that for any collection of non-empty sets, a simultaneous selection of one element from each set exists. Its distinguishing feature among the axioms of ZF is that it asserts existence without specifying any rule of selection [2]. This non-constructive character is a diagnostic signal: when AC is required to establish a property of a mathematical structure, the requirement indicates the structure lacks the canonical geometric information that would make the selection determinate. AC substitutes existence for construction, and in doing so marks the precise points where geometric specification is absent.

The following survey identifies at each level of the standard number hierarchy the specific geometric information whose absence necessitates AC. This is not a critique of the constructions — they are all correct and complete for their algebraic purposes. It is a map of what the algebraic constructions leave uncharacterized, and therefore what the Möbius connection supplies.

2.2 The Integers: The Orientation Choice

The standard construction of $\mathbb{Z}$ from $\mathbb{N}$ proceeds via equivalence classes of pairs: a Peano integer is defined as a pair $(n, m) \in \mathbb{N} \times \mathbb{N}$ with $(n, m) \sim (n', m')$ iff $n + m' = n' + m$ [5]. This construction requires *two* natural numbers to represent *one* integer, with an equivalence relation to collapse the redundancy. The designation of which direction is “positive” is a choice: the circle S^1 from which winding numbers generate $\mathbb{Z}$ has two orientations, and neither is privileged by the circle’s intrinsic geometry.

Proposition 2.1 (Orientation choice for $\mathbb{Z}$). *The standard construction of $\mathbb{Z}$ from the loop space $\pi_1(S^1) \cong \mathbb{Z}$ requires a choice of basepoint orientation on S^1 . This choice is not determined by the topological structure of S^1 and constitutes an implicit invocation of AC in the infinite case.*

Proof. The two orientations of S^1 are exchanged by the orientation-reversing homeomorphism $\theta \mapsto -\theta$. No intrinsic feature of S^1 distinguishes them: the symmetry group $O(2)$ acts transitively on the set of orientations. To declare one “positive” requires selecting an element from a two-element set with no canonical distinguishing property. $\square$

Remark 2.2. *The Peano induction axiom encodes the orientation choice asymmetrically: the successor operation runs in one direction. Properties of negative integers require separate treatment, reflecting the absence of a symmetric generative mechanism.*

2.3 The Reals: Three Invocations

The construction of $\mathbb{R}$ from $\mathbb{Q}$ requires AC at three distinct points.

(i) **Ultrafilter construction.** One standard construction of $\mathbb{R}$ via non-standard analysis uses a non-principal ultrafilter on $\mathbb{N}$ [6]. The existence of a non-principal ultrafilter requires AC (specifically, the Boolean Prime Ideal Theorem) [7].

(ii) **Well-ordering.** AC is equivalent to the well-ordering principle [3]. No explicit well-ordering of $\mathbb{R}$ can be constructed in ZF; its existence is purely existential.

(iii) **Non-measurable sets.** Vitali's construction uses AC to select one representative from each equivalence class of reals differing by a rational [8]. The resulting set is Lebesgue non-measurable.

Proposition 2.3 (AC dependence of $\mathbb{R}$). *Each of the following is unprovable in ZF without AC: (a) existence of a non-principal ultrafilter on $\mathbb{N}$; (b) existence of a well-ordering of $\mathbb{R}$; (c) existence of a Lebesgue non-measurable subset of $[0, 1]$.*

2.4 The Complex Numbers: Orientation Left Unspecified

Theorem 2.4 (Steinitz–AC, 1910). *The existence and uniqueness (up to isomorphism) of the algebraic closure of an arbitrary field requires AC. Without AC, a field may have two non-isomorphic algebraic closures [4].*

Remark 2.5 (Scope of the Steinitz theorem). *The Steinitz theorem applies to algebraic closures of arbitrary fields. The specific case $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$ is constructible in ZF without AC, and the Fundamental Theorem of Algebra — that every polynomial over $\mathbb{C}$ has a root — is provable without AC. The non-canonicity of $\mathbb{C}$ does not arise from the existence or uniqueness of the algebraic closure of $\mathbb{R}$ as an abstract field.*

It arises from a different and subtler source: the orientation of $\mathbb{C}$ is underdetermined by the algebraic construction. The quotient $\mathbb{R}[x]/(x^2 + 1)$ produces two roots of $x^2 + 1 = 0$, related by complex conjugation, with no algebraic means of distinguishing them. Selecting which root is called i — determining which direction is “positive imaginary” and thereby fixing the embedding $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$ — requires a geometric datum that the algebraic construction does not provide.

It is this orientation gap, not the existence of $\mathbb{C}$ itself, that the Möbius connection resolves.

Corollary 2.6 (Orientation underdetermination of $\mathbb{C}$). *The algebraic construction $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$ does not canonically determine the orientation of $\mathbb{C}$: the two roots of $x^2 + 1 = 0$ are algebraically indistinguishable, and selecting which is i requires a choice not supplied by the algebra. Claims about the arithmetic significance of specific positions in $\mathbb{C}$ — such as the assertion that zeros lie on a specific locus defined by $\operatorname{Re}(s)$ — presuppose a canonical orientation that the algebraic construction does not provide.*

Remark 2.7 (The orientation consequence). *The orientation of $\mathbb{C}$ is not determined by the algebraic construction. The embedding $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$ — which fixes what “positive imaginary” means and thereby gives $\operatorname{Re}(s)$ its meaning — is supplied by convention in the standard treatment. Claims about the arithmetic significance of specific positions in $\mathbb{C}$ (such as the assertion that zeros of ζ lie on $\operatorname{Re}(s) = \frac{1}{2}$) presuppose this orientation is canonically fixed. A canonical generator of $\mathbb{C}$'s orientation from $\mathbb{R}$ is required for such claims to be invariant under change of presentation — not merely true, but meaningfully and canonically true.*

Remark 2.8 (On the indistinguishability of i and $-i$). *The field $\mathbb{C}$ admits a non-trivial field automorphism: complex conjugation $\sigma : z \mapsto \bar{z}$, which swaps i and $-i$. This means that i and $-i$ are algebraically indistinguishable within $\mathbb{C}$ as an abstract field: no field-theoretic statement can tell them apart. This fact is well-known and is a direct consequence of the construction of $\mathbb{C}$ as $\mathbb{R}[x]/(x^2 + 1)$, which produces two roots $\pm x$ of $x^2 + 1 = 0$ with no internal means of distinguishing them.*

The standard response is that this symmetry is harmless: any statement invariant under complex conjugation (such as the assertion that all non-trivial zeros of $\zeta(s)$ lie on $\text{Re}(s) = \frac{1}{2}$) is well-posed regardless.

This response is correct but incomplete. It shows that the zero set of ζ is well-defined. It does not show that the critical line — defined as the locus $\text{Re}(s) = \frac{1}{2}$ relative to the embedding of $\mathbb{R}$ into $\mathbb{C}$ — is canonically identified. The real part of a complex number is not an intrinsic property of a field element; it is a property relative to a chosen embedding $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$. Two presentations of $\mathbb{C}$ related by the automorphism σ agree on the zero set of ζ but assign opposite imaginary signs to every non-real zero. The statement “this zero lies on $\text{Re}(s) = \frac{1}{2}$ ” is therefore invariant under σ , but only because both s_0 and $\bar{s}_0$ lie on that line simultaneously — meaning the symmetry axis $\text{Re}(s) = \frac{1}{2}$ is self-conjugate, not because the position of any individual zero is conjugation-independent.

The deeper question is what fixes the embedding ι in the first place. In the standard construction, ι is supplied by the orientation of $\mathbb{R}$, which is itself fixed by the convention that positive integers come first in the Peano successor — a stipulation, not a derivation. The sign of $\sqrt{-1}$ is then fixed relative to this orientation. But the drawing convention is not the mathematics.

What the Möbius connection provides is not a drawing convention but a geometric derivation: the imaginary unit is the holonomy generator of the connection, determined by which direction of traversal the connection encodes. The sign is fixed by the structural asymmetry between one traversal (which returns -1) and two traversals (which return $+1$). This asymmetry is not a choice; it is the content of the $\mathbb{Z}_2$ holonomy.

The relationship between the standard algebraic $\mathbb{C}$ and the canonical $\mathbb{C}$ generated by the Möbius connection is analogous to the relationship between a topological manifold and a Riemannian manifold. The topological manifold is correct and complete for topology. But to understand geodesics, curvature, and parallel transport, you need the metric. The metric does not fix a flaw in topology; it completes a structure that topology leaves uncharacterized. The Möbius connection stands in the same relation to the algebraic $\mathbb{C}$: not a repair, but a geometric completion.

3 The Möbius Connection

3.1 Definition and Basic Properties

Definition 3.1 (Möbius connection). *The Möbius connection is the $U(1)$ connection form*

$$\omega = \frac{1}{2} d\theta$$

on the circle $S^1 = \{e^{i\theta} : \theta \in [0, 2\pi)\}$, equipped with the line bundle $L \rightarrow S^1$ with structure group $U(1)$.

The holonomy of ω around the full circle is:

$$H(\gamma) = \exp\left(\oint_{S^1} \omega\right) = e^{i\pi} = -1.$$

One traversal returns -1 ; two traversals return $+1$. The connection defines a non-trivial $\mathbb{Z}_2$ holonomy: the bundle L is the Möbius bundle, and its double cover is the first non-trivial covering space of S^1 .

3.2 Uniqueness: The Connection-Theoretic Argument

Consider the family of flat connections $\omega_\alpha = \alpha d\theta$ for $\alpha \in \mathbb{R}$. The holonomy is $H_\alpha = e^{2\pi i \alpha}$. This family partitions into classes with distinct structural characters.

Definition 3.2 (Three intrinsic properties). *A flat connection ω_α on S^1 is:*

- (i) Discrete if $\text{im}(\rho_\alpha)$ is a finite subgroup of $U(1)$ — equivalently, $\alpha \in \mathbb{Q}$.
- (ii) Irreducible if $\text{im}(\rho_\alpha)$ has prime order — equivalently, $\alpha = 1/p$ for a prime p .
- (iii) Involutive if $H_\alpha^2 = 1$ — equivalently, the holonomy squares to the identity.

These are not axioms designed to select a predetermined answer. They are standard mathematical attributes — finiteness, primality, and involutivity — applied to the holonomy representation of a flat connection. Each corresponds to a distinct structural criterion:

Discreteness separates connections that impose structure at a definite scale from those whose holonomy is dense in $U(1)$. An irrational holonomy generates a dense orbit with no periodicity — it cannot distinguish any finite set of points on the fiber from any other. A connection with continuous ambiguity of this kind cannot be fundamental: it can be approximated arbitrarily closely by other connections without changing its qualitative character. It says “I am not fundamental.”

Irreducibility separates connections whose holonomy group is an atomic unit from those that decompose. A composite holonomy $\mathbb{Z}_n$ with n composite factors as $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$, carrying internal structure beyond what traversal requires. Such a connection says “I can still be expressed as a composite.” Irreducibility is primality applied to the holonomy group: the connection contributes exactly one layer of structure, not a product of simpler layers.

Involutivity matches the topological content of orientation. A manifold is orientable or not. The orientation-reversing operation, applied twice, returns to the starting orientation. This is not a convention — it is the mathematical content of what “orientation” means. A connection whose holonomy has order $p > 2$ imposes a p -fold cyclic structure that goes beyond orientation: it distinguishes p directions rather than two. Orientation is binary, and only an involutive holonomy captures this.

Theorem 3.3 (Uniqueness of the Möbius connection). *The connection $\omega = \frac{1}{2}d\theta$ is the unique element of $\{\omega_\alpha : \alpha \in \mathbb{R}\}$ that is simultaneously discrete, irreducible, and involutive.*

Proof. The finite cyclic subgroups of $U(1)$ are $\mathbb{Z}_n$ for $n \geq 1$. Among these, the irreducible (prime-order) subgroups are $\mathbb{Z}_p$ for primes p . Among those, the involutive subgroups ($H^2 = 1$) require the order to divide 2. The only prime dividing 2 is $p = 2$. Therefore $\text{im}(\rho_\alpha) = \mathbb{Z}_2 = \{1, -1\}$, which forces $\alpha = 1/2$ and $\omega = \frac{1}{2}d\theta$. $\square$

Remark 3.4 (Field realizability as consequence). *The fact that $\mathbb{Z}_2$ holonomy generates a field extension of $\mathbb{R}$ (specifically $\mathbb{R}[j]/(j^2 + 1) \cong \mathbb{C}$), while $\mathbb{Z}_n$ for $n \geq 3$ generates only a ring with zero divisors, is a consequence of the uniqueness, not its cause. The connection is canonical before one knows it generates a field. That it does generate a field is an independent verification of the selection, not a condition imposed to achieve it.*

Remark 3.5 (Two independent routes to canonicity). *The connection-theoretic argument (Theorem 3.3) and the self-grounding argument (developed in the companion papers) are independent routes to the same object. The former uses only intrinsic properties of flat connections; the latter uses the categorical fixed-point structure of $\mathbb{Z}_2\text{-Set}$. That two independent arguments converge on $\omega = \frac{1}{2}d\theta$ is stronger evidence for canonicity than either alone.*

Remark 3.6. *The identification $j^2 = -1$ is not a definition imposed from outside: it is the holonomy condition. Two traversals return to identity: $(-1)^2 = 1$. The imaginary unit i is what the holonomy generator is.*

3.3 Categorical Formalization: The Category OAS

Theorem 3.3 establishes the Möbius connection as the unique flat connection on S^1 with discrete, irreducible, involutive holonomy. This section elevates the uniqueness to a *universal property* by constructing a category whose initial object is (S^1, ω) . The category formalizes the already-established selection; it does not perform the selection.

The four axioms below codify the properties that the connection was already shown to possess. Each corresponds to one of the connection-theoretic criteria or their algebraic consequences: nontrivial holonomy (from discreteness), finite closure (from discreteness), minimality (from irreducibility), and field realizability (the algebraic consequence of involutive $\mathbb{Z}_2$ holonomy, established in Remark 3.4).

The four conditions

Condition (A1): Nontrivial holonomy. The holonomy representation $\rho_\omega : \pi_1(X) \rightarrow U(1)$ is nontrivial. This is the algebraic expression of discreteness: a trivial holonomy carries no orientational record.

Condition (A2): Finite closure. $\text{im}(\rho_\omega)$ is finite. The covering space generated by the connection is finite-sheeted. This is the algebraic expression of discreteness combined with closure: the generated structure eventually closes.

Condition (A3): Minimality. No proper quotient of $\text{im}(\rho_\omega)$ satisfies both (A1) and (A2). This is the algebraic expression of irreducibility: the holonomy group contributes exactly one layer of structure.

Condition (A4): Field realizability. The algebra $\mathbb{R}[\text{im}(\rho_\omega)]$ contains a field extension of $\mathbb{R}$. This is the algebraic consequence of the holonomy being $\mathbb{Z}_2$: the ring $\mathbb{R}[\mathbb{Z}_2] \cong \mathbb{C}$ is a field, while $\mathbb{R}[\mathbb{Z}_n]$ for $n \geq 3$ contains zero divisors.

The category and its initial object

The four conditions jointly characterize the class of geometric structures whose holonomy is discrete, irreducible, involutive, and algebraically productive. We codify this as a category.

Definition 3.7 (Oriented Arithmetic Structure). *An oriented arithmetic structure is a triple (X, L, ω) where X is a connected 1-manifold with $\pi_1(X) \cong \mathbb{Z}$ (which forces $X \simeq S^1$ up to homotopy), $L \rightarrow X$ is a principal $U(1)$ -bundle, and ω is a $U(1)$ -connection on L , satisfying:*

(A1) Nontrivial holonomy: $\rho_\omega : \pi_1(X) \rightarrow U(1)$ is nontrivial.

(A2) Finite closure: $\text{im}(\rho_\omega)$ is finite.

(A3) Minimality: no proper quotient of $\text{im}(\rho_\omega)$ satisfies both (A1) and (A2).

(A4) Field realizability: the algebra $\mathbb{R}[\text{im}(\rho_\omega)]$ contains a field extension of $\mathbb{R}$.

A morphism $f : (X, \omega) \rightarrow (Y, \eta)$ is a smooth map $f : X \rightarrow Y$ inducing a π_1 -homomorphism, together with a bundle map that satisfies $f^*\eta = \omega$. This condition forces morphisms to preserve holonomy data exactly: no reparametrization of orientation is permitted. We denote this category **OAS**.

Remark 3.8. The four conditions (A1)–(A4) codify the connection-theoretic properties already established in Theorem 3.3 and Remark 3.4. Conditions (A1)–(A3) are direct translations of discreteness, finite closure, and irreducibility into the language of holonomy representations. Condition (A4) is the algebraic consequence that $\mathbb{Z}_2$ holonomy generates a field. The category **OAS** does

not select the Möbius connection; it provides the categorical habitat in which the already-selected connection has a universal property.

Theorem 3.9 (Möbius connection as initial object). *In the category **OAS**, the object (S^1, ω_M) with $\omega_M = \frac{1}{2}d\theta$ is an initial object: for every object $(X, \omega) \in \mathbf{OAS}$, there exists a unique morphism $(S^1, \omega_M) \rightarrow (X, \omega)$.*

Proof. Let (X, ω) be any object of **OAS**. Since X is a connected 1-manifold with $\pi_1(X) \cong \mathbb{Z}$, we have $X \simeq S^1$. The holonomy representation $\rho_\omega : \mathbb{Z} \rightarrow U(1)$ is determined by the image of the generator $1 \in \mathbb{Z}$; write $\rho_\omega(1) = e^{2\pi i \alpha}$ for some $\alpha \in \mathbb{R}$.

By (A2), $\text{im}(\rho_\omega)$ is finite, so $e^{2\pi i \alpha}$ has finite order, which forces $\alpha \in \mathbb{Q}$.

By (A3), $\text{im}(\rho_\omega)$ is minimal among finite subgroups of $U(1)$ satisfying (A1). The finite subgroups of $U(1)$ are exactly the cyclic groups $\mathbb{Z}_n$ for $n \geq 1$. The minimal nontrivial cyclic subgroup is $\mathbb{Z}_2 = \{1, -1\}$.

By (A4), $\mathbb{R}[\text{im}(\rho_\omega)]$ must contain a field extension of $\mathbb{R}$. For $\text{im}(\rho_\omega) = \mathbb{Z}_n$ with $n \geq 3$, the algebra $\mathbb{R}[\mathbb{Z}_n]$ contains zero divisors and is not a field. Only $\mathbb{Z}_2$ generates $\mathbb{R}[\mathbb{Z}_2] = \mathbb{R}[j]/(j^2 + 1) \cong \mathbb{C}$, which is a field.

Therefore $\rho_\omega(1) = -1$, which forces $\alpha = 1/2$ and $\omega = \frac{1}{2}d\theta = \omega_M$.

The unique morphism $(S^1, \omega_M) \rightarrow (X, \omega)$ is now determined: since $\pi_1(S^1) \cong \mathbb{Z}$ is the free group on one generator, a π_1 -homomorphism from S^1 to $X \simeq S^1$ is determined by where the generator maps. The morphism condition $f^*\eta = \omega$ fixes this: the generator of $\pi_1(S^1)$ must map to the generator of $\pi_1(X)$ in the direction that identifies their holonomies. Since both holonomies are -1 (forced by the argument above), this identification is unique.

Uniqueness of the morphism follows from the freeness of $\pi_1(S^1) \cong \mathbb{Z}$: a homomorphism from a free group on one generator is determined by the image of that generator, and the morphism condition leaves no freedom in choosing that image. $\square$

Remark 3.10 (Robustness under reformulation). *The conclusion of Theorem 3.9 is robust under reasonable variations of the category definition. If (A4) is replaced by the weaker requirement that $\mathbb{R}[\text{im}(\rho_\omega)]$ be a division algebra (rather than a field), the conclusion is unchanged: the only finite subgroup of $U(1)$ generating a real division algebra is $\mathbb{Z}_2$, since higher cyclic groups produce zero divisors. If the structure group is changed from $U(1)$ to any compact abelian Lie group, the same argument forces the minimal nontrivial closed subgroup satisfying (A3) and (A4) to be $\mathbb{Z}_2$. The Möbius connection is not an artifact of the specific formulation of **OAS**; it is the unique survivor of the constraint pressure the four derived conditions impose.*

Remark 3.11 (On the use of $\mathbb{R}$ in the OAS definition). *Conditions (A4) and the uniqueness proof both invoke $\mathbb{R}$ as the base field over which the generated algebra is evaluated. This could appear circular: the paper derives $\mathbb{R}$ from the Möbius connection, yet uses $\mathbb{R}$ to state the conditions that select the Möbius connection.*

The circularity is apparent rather than real. The OAS framework uses $\mathbb{R}$ in the same way that Euclidean geometry uses the notion of a point, or that Lawvere's natural number object uses an ambient category: as a working vocabulary that the construction will eventually account for from within, but which serves as a provisional starting point. The justification is coherentist rather than foundationally linear.

Specifically, the $\mathbb{R}$ deployed in (A4) is the pre-theoretic real line, treated as a given ordered field with its standard properties. The $\mathbb{R}$ derived from the Möbius connection (§4) is the fixed-point locus of the connection's conjugation involution, shown to have the Dedekind completeness property (§4.3). The non-circularity claim is that these two objects are the same: the pre-theoretic $\mathbb{R}$ has

exactly the properties that the derived fixed-point locus possesses, and no additional structure was assumed in (A4) that the derived $\mathbb{R}$ fails to exhibit.

This is the standard coherentist pattern in foundational mathematics. Frege's and Russell's constructions of arithmetic proceed similarly: one uses informal arithmetic to state the conditions that formal arithmetic must satisfy, then verifies that the formal construction meets them. The validity of the construction is demonstrated by conservation, not by linear independence from all prior vocabulary.

Remark 3.12 (The extension constraint: a proof sketch). *Beyond the internal argument of Theorem 3.9, the Möbius connection is distinguished by an external constraint that any competing canonical generator must satisfy: it must extend coherently through the normed division algebra tower. We sketch why the Möbius connection satisfies this constraint, to the degree needed to motivate the discussion of higher-level structure in §6.*

The Cayley-Dickson doubling rule. The rule $(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c})$ doubles a normed division algebra A into a new algebra $A \oplus A$. At each stage, the rule introduces a new generator j satisfying $j^2 = -1$ and the anti-involution $(a, \bar{b}) = (\bar{a}, -b)$.

The holonomy interpretation. In the Möbius framework, each new generator introduced by doubling is a new holonomy generator: a direction of traversal that returns -1 after one circuit. The doubling rule's minus sign $(-\bar{d}b)$ is the algebraic encoding of the orientation flip that the holonomy produces. At the first doubling ($\mathbb{R} \rightarrow \mathbb{C}$), the single generator i with $i^2 = -1$ is the Möbius holonomy. At the second doubling ($\mathbb{C} \rightarrow \mathbb{H}$), the new generator j satisfies $j^2 = -1$, and $k = ij$ is the third imaginary unit of the quaternions. The key structural fact is that $\mathbb{H}$ contains exactly three canonical copies of $\mathbb{C}$: the planes spanned by $\{1, i\}$, $\{1, j\}$, $\{1, k\}$. Each is generated by its own holonomy element. The base copy $\{1, i\}$ is the canonical $\mathbb{C}$ of this paper; the other two are distinct orientations of the same structure, embedded at the $\mathbb{H}$ level.

The failure modes are holonomy failures. Commutativity fails at $\mathbb{H}$ ($ij \neq ji$) because two independent holonomy generators do not commute under composition. Associativity fails at $\mathbb{O}$ because three independent holonomy generators do not associate. The division algebra property fails at the sedenion level (dimension 16) because the Cayley-Dickson doubling at that stage introduces zero divisors: the holonomy encoding capacity of the doubling rule is exceeded by the underlying topological complexity. These failures are not accidents of the construction; they follow from properties of the sphere-fibration structure that the holonomy generates at each level, as established by Adams [1].

Significance for this paper. No known alternative construction of $\mathbb{C}$ from $\mathbb{R}$ reproduces these nested holonomy levels at precisely the right dimensions (1, 2, 4, 8) with the right failure modes. This is the sense in which extension coherence is evidence for canonicity: the construction does not merely generate $\mathbb{C}$, it generates an entire hierarchy of structures whose properties follow from the same holonomy mechanism. The nested structure of $\mathbb{C}$ -copies within $\mathbb{H}$ and $\mathbb{O}$ is what the companion paper addresses when establishing that non-canonical $\mathbb{C}$ -copies are reached by holonomy operations from the canonical $\mathbb{C}$, and that analytic continuation on $\mathbb{C}$ does not encounter these operations. A full derivation of the tower from the Möbius connection is reserved for separate work.

3.4 Successor Generation vs. Loop Generation

The Peano successor operation $S : n \mapsto n + 1$ is intrinsically oriented: there is a built-in forward direction, and negative integers are introduced afterward as formal inverses or equivalence classes. Orientation is stipulated at generation time, not derived.

The loop-based construction is symmetric. The circle S^1 has $\pi_1(S^1) \cong \mathbb{Z}$ without a preferred

orientation: clockwise and counterclockwise are initially indistinguishable. Direction is absent until supplied by additional geometric structure.

Proposition 3.13 (Successor arithmetic as gauge-fixed loop arithmetic). *The Peano successor construction is a gauge-fixed presentation of loop-based integer generation: it corresponds to choosing a global orientation of S^1 before generating $\mathbb{Z}$. This choice is not forced by the loop structure and constitutes the orientation choice of Proposition 2.1.*

Under the Möbius connection, orientation is replaced by a structural fact: positive and negative winding are distinguished by holonomy eigenvalue. The Peano pair construction (n, m) representing $n - m$ implicitly encodes the two sheets of the Möbius double cover; the equivalence relation $\sim$ collapses the redundancy of describing the same displacement from two sheet-perspectives.

3.5 The Cauchy–Riemann Equations as Co-definition Consistency

The Möbius connection distinguishes two structurally different directions on $\mathbb{C}$: the real axis (traversal direction, accumulating holonomy) and the imaginary axis (orientation-tracking direction, perpendicular to traversal). A function on $\mathbb{C}$ that treats these as structurally independent is precisely an analytic function.

Proposition 3.14 (Analyticity as connection consistency). *A differentiable function $f : \mathbb{C} \rightarrow \mathbb{C}$ preserves the distinction between the traversal and orientation-tracking directions of the Möbius connection if and only if f satisfies the Cauchy–Riemann equations.*

Analytic functions are not an additional condition imposed on $\mathbb{C}$: they are the functions respecting the structural content of the connection from which $\mathbb{C}$ was derived.

3.6 Implications for Complex Analysis

The preceding sections establish that the Möbius connection generates $\mathbb{C}$ canonically. This subsection makes explicit what that changes for complex analysis — not as a correction of existing results, but as a structural completion that makes certain problems accessible that were previously opaque.

Standard complex analysis operates on a geometrically silent $\mathbb{C}$

In the standard framework, $\mathbb{C}$ is algebraically complete: every polynomial has a root, the field structure is well-defined, and the tools of analysis — holomorphic functions, contour integration, analytic continuation — are fully operational. Nothing in this framework is wrong.

But the framework is geometrically silent. The internal constitution of $\mathbb{C}$ — which directions are structurally distinguished, how the real and imaginary axes are related to each other, what geometric content the embedding $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$ carries — is unspecified. The real axis is identified by convention (the fixed locus of conjugation), the imaginary axis is its orthogonal complement, and the critical line $\operatorname{Re}(s) = \frac{1}{2}$ is defined relative to these conventional choices.

Analytic continuation, the central tool of complex analysis, extends functions along paths in $\mathbb{C}$ without knowing which paths are structurally distinguished and which are not. It works “blind”: every path is as good as any other, and the function’s behavior along one path constrains its behavior along all others. This universality is the power of analytic continuation, but it is also its limitation. When the question is *why a function’s zeros lie on a specific locus*, analytic continuation can verify that they do (by checking the function’s values along the locus) but cannot explain *why that locus and no other* — because the locus has no canonical status in the framework.

This is why the Riemann Hypothesis is hard. The critical line $\text{Re}(s) = \frac{1}{2}$ has no canonical status in a $\mathbb{C}$ whose orientation is conventional. It is defined relative to an embedding $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$ that the algebraic construction does not provide. Every proof attempt must work around this by exploiting properties of ζ that happen to be invariant under conjugation — which succeeds in making the *statement* well-posed, but means the proofs never engage with *why the critical line is special*, only with the fact that ζ 's symmetries happen to respect it.

Canonical complex analysis operates on a geometrically specified $\mathbb{C}$

The Möbius connection makes the internal geometry of $\mathbb{C}$ explicit. The two-track generation (Section 4) establishes:

1. The real axis is Track 1's completion — the additive integer structure completed to a continuum. It is the fixed-point locus of complex conjugation $z \mapsto \bar{z}$, carrying trivial holonomy.
2. The critical line $\text{Re}(s) = \frac{1}{2}$ is Track 2's completion — the multiplicative prime structure completed to a continuum. It is the fixed-point locus of the canonical involution $s \mapsto 1 - s$, carrying the imaginary holonomy structure generated by the prime logarithms.
3. These two loci are orthogonal because the two tracks are independent (guaranteed by the fundamental theorem of arithmetic). Their coexistence generates $\mathbb{C}$.

In this framework, the critical line is not a conventional locus defined relative to an arbitrary embedding. It is the fixed-point set of a canonical involution determined by the connection — the unique locus where the two monodromy sheets of the Möbius double cover balance. Its position at $\text{Re}(s) = \frac{1}{2}$ is forced by the connection's geometry, not chosen by convention.

Analytic continuation in the canonical $\mathbb{C}$ becomes “sighted”: it extends functions along paths that carry structural meaning relative to the connection. The Cauchy–Riemann equations (Proposition 3.14) are not merely a regularity condition but a *consistency condition with the connection*. A function that is analytic in the canonical $\mathbb{C}$ automatically respects the two-track structure — it cannot “see” one track without “seeing” the other, because both tracks are generated by the same holonomy mechanism.

Consequences for the characteristic class hierarchy

The connection between the canonical $\mathbb{C}$ and the standard tools of algebraic topology deserves explicit statement. The standard framework's “choices” — which cohomology theory, which characteristic class, which coefficient ring — are not arbitrary preferences. They are structural accommodations for the ambiguity that arises from working with the full continuous family of connections on S^1 (or with no specific connection at all), rather than fixing the canonical connection first.

Each level of the normed division algebra tower generates a specific type of ambiguity, and each characteristic class manages one level:

1. **Stiefel–Whitney classes** manage the $\mathbb{Z}/2$ ambiguity at the $\mathbb{C}$ level — the orientation choice that the Möbius connection resolves.
2. **Chern classes** manage the S^2 explosion at $\mathbb{H}$ — the proliferation of complex structures within the quaternionic framework.
3. **Pontryagin classes** manage the real reduction back from $\mathbb{H}$ to $\mathbb{R}$ — the forgetful functor that erases quaternionic structure.
4. **Stable homotopy** manages the zero-divisor explosion beyond $\mathbb{O}$ — the constraint geometry of the post-NDA tower.

Each of these is re-deriving locally what the canonical connection establishes globally. The free resolutions underlying the Tor functor are covering maps of S^1 , which means they already implicitly

reference connection structure. The Universal Coefficient Theorem tracks how much structure is lost when projecting from one level’s explosion to another’s — it is the accounting system for the cascade of accommodation costs that the tower generates.

With the canonical connection fixed, these tools acquire a reference point: Tor measures distortion relative to a *known* canonical form, not relative to an unspecified background. The “choices” of algebraic topology become determinate consequences of where one sits in the tower, not independent parameters to be selected.

What this enables

The companion paper (Paper II) exploits this geometric completion directly. Because the critical line is canonically determined — as the fixed-point locus of the connection’s involution, not as a conventional line defined by an arbitrary embedding — the statement “all non-trivial zeros of $\zeta(s)$ lie on $\operatorname{Re}(s) = \frac{1}{2}$ ” becomes a statement about the connection’s geometry rather than about the analytic properties of a specific function. The proof strategy shifts from asking “does ζ happen to vanish only on this line?” to asking “can the canonical two-track section of the Möbius connection vanish anywhere else?” — a question that the connection’s structure is equipped to answer.

This is the sense in which the canonical generation of $\mathbb{C}$ is not merely a cleaner foundation for existing results, but an *enlargement of what complex analysis can reach*. Problems that depend on the canonical status of specific loci — problems that the geometrically silent framework cannot even formulate as geometric questions — become accessible once the silence is filled.

Remark 3.15 (Geometric priority). *A pattern visible throughout this paper, and increasingly explicit as the series progresses, deserves naming: algebraic structure is downstream of geometric constitution. The algebraic construction of $\mathbb{C}$ produces a correct field whose geometric content is unspecified. The Möbius connection specifies that content, and the algebraic properties — analyticity, the Cauchy–Riemann equations, the two-track structure, the characteristic class hierarchy — follow as consequences. The algebra does not generate the geometry; the geometry generates the algebra.*

This is not a philosophical preference imposed on the mathematics. It is the structure of the argument: at every point where the algebraic framework requires a choice (orientation of $\mathbb{Z}$, embedding $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$, selection of characteristic classes), the Möbius connection supplies a geometric determination that eliminates the choice. The relationship is asymmetric: geometry resolves algebraic ambiguity, but algebra does not resolve geometric silence.

The standard practice of treating algebra and geometry as independent domains that happen to interact is not wrong — it is incomplete in the same way that the algebraic $\mathbb{C}$ is incomplete. The interaction has a direction: geometric structure is logically prior, and algebraic structure is its expression in a specific representational mode. The subsequent papers in this series provide further evidence for this priority, culminating in the constraint geometry of the Cayley–Dickson tower (Paper IV), where the post-algebraic structure is entirely geometric in character.

4 Repair of the Orientation Gap

4.1 The Integers: Holonomy Replaces Choice

Theorem 4.1 (Canonical orientation of $\mathbb{Z}$). *The integers constructed as winding numbers of the Möbius-equipped S^1 carry a canonical orientation: positive and negative integers are distinguished by the holonomy accumulated under the connection, without any choice of basepoint orientation.*

Proof. Under $\omega = \frac{1}{2}d\theta$, the holonomy of a path winding n times is $e^{in\pi} = (-1)^n$. Positive and negative winding classes are separated by the holonomy eigenvalue. The distinction is encoded in the connection's response to traversal direction and requires no choice. $\square$

Remark 4.2 (Two structures on $\mathbb{Z}$ from a single generation). *The integers generated by traversal carry two structures simultaneously, both consequences of the same holonomy mechanism.*

Additive structure. *Each integer n is a winding number — a count of oriented traversals. Addition is concatenation of paths. The holonomy $(-1)^n$ distinguishes positive from negative winding. This is the structure that generates the ordering and the sign of $\mathbb{Z}$.*

Multiplicative structure. *Among the positive integers, certain elements are multiplicatively irreducible: they cannot be expressed as the product of two smaller positive integers. These are the prime numbers. In the traversal picture, the primes are the irreducible traversal steps — the winding classes that cannot be decomposed into shorter windings in the multiplicative sense. Every positive integer is a unique product of primes (the fundamental theorem of arithmetic), which means every additive winding class has a unique decomposition into irreducible multiplicative steps.*

The crucial point is that both structures emerge from the same traversal process. The primes are not external inputs to the framework — they are identified as the irreducible elements of the multiplicative structure that the holonomy generation of $\mathbb{Z}$ produces. The holonomy accumulated by traversing k distinct prime steps is $(-1)^k$, which is the definition of the arithmetic Möbius function $\mu(n) = (-1)^k$ for squarefree n with k prime factors.

This two-structure character of $\mathbb{Z}$ is not incidental. It is the foundation of the two-track structure that generates $\mathbb{C}$: the additive structure (Track 1) and the multiplicative prime structure (Track 2) are two aspects of one object, unified by the fundamental theorem of arithmetic. Their interaction generates the full canonical number hierarchy.

4.2 The Rationals: The Two-Track Pivot

The rational numbers $\mathbb{Q}$ arise from $\mathbb{Z}$ by admitting multiplicative inverses: $\mathbb{Q}$ is the field of fractions of $\mathbb{Z}$. In the Möbius framework this step is more significant than it first appears, because it is where the two structures of $\mathbb{Z}$ — additive and multiplicative — develop into two independent but consistent tracks that together generate $\mathbb{C}$.

Track 1 (additive). The field of fractions construction gives $\mathbb{Q}$ as ratios p/q of integers. This extends the additive integer structure with multiplicative inverses. The result is an ordered field, dense in $\mathbb{R}$, generated by rational subdivisions of the unit interval. Every rational is reachable from $\mathbb{Z}$ by a finite sequence of additions and divisions.

Track 2 (multiplicative). The positive rationals $\mathbb{Q}_+$ under multiplication form a group isomorphic to the free abelian group on countably many generators — and those generators are exactly the primes. Every positive rational p/q factors uniquely as $\prod_i p_i^{a_i}$ for integers a_i (positive for primes in the numerator, negative for primes in the denominator). This is the fundamental theorem of arithmetic extended to $\mathbb{Q}_+$.

The logarithm map converts this multiplicative structure into an additive one: $\log : \mathbb{Q}_+ \rightarrow \mathbb{R}$ sends $\prod_i p_i^{a_i}$ to $\sum_i a_i \log p_i$. The image $\log(\mathbb{Q}_+)$ is the $\mathbb{Z}$ -linear span of $\{\log p : p \text{ prime}\}$ — the set of all integer linear combinations of prime logarithms. This set is countably infinite, dense in $\mathbb{R}$, and its elements are linearly independent over $\mathbb{Q}$ (since no prime is a rational power of another prime, by the fundamental theorem of arithmetic).

Proposition 4.3 (Two dense structures on $\mathbb{R}$ from $\mathbb{Q}$). *The rational numbers generate two distinct dense subsets of $\mathbb{R}$:*

(i) *The rationals $\mathbb{Q}$ themselves, dense in $\mathbb{R}$ via additive approximation (Track 1).*

(ii) *The set $\log(\mathbb{Q}_+) = \bigoplus_p \mathbb{Z} \cdot \log p$, dense in $\mathbb{R}$ via prime-logarithm combinations (Track 2).*

Both are generated by the same integers via the fundamental theorem of arithmetic. Their Dedekind completions are both copies of $\mathbb{R}$, but embedded orthogonally in $\mathbb{C}$: Track 1 completes to the real axis, and Track 2 completes to the imaginary direction whose fixed locus under the canonical involution is the critical line $\operatorname{Re}(s) = \frac{1}{2}$.

Proof sketch. That $\mathbb{Q}$ is dense in $\mathbb{R}$ is standard. That $\log(\mathbb{Q}_+)$ is dense in $\mathbb{R}$ follows from the linear independence of $\{\log p\}$ over $\mathbb{Q}$ together with the density of $\mathbb{Z}$ -linear combinations of any infinite linearly independent set. The orthogonality of the two completions in $\mathbb{C}$ follows from the structure of $\mathbb{C}$ established in Theorem 4.8: the real axis is the fixed-point locus of $z \mapsto \bar{z}$, and the imaginary direction is the orthogonal complement generated by the holonomy operator i . $\square$

Remark 4.4 (The fundamental theorem of arithmetic as the unifying theorem of the two tracks). *The fundamental theorem of arithmetic — that every positive integer has a unique prime factorization — is the theorem that makes Track 1 and Track 2 consistent with each other. It guarantees that the additive description (integer sums) and the multiplicative description (prime products) are two encodings of the same structure, not two independent structures that happen to coexist.*

This consistency is what allows $\mathbb{C}$ to be a field rather than merely a product of two real lines. If the two tracks were independent, their completions would generate an unrelated pair of real lines. Because they are the same structure described two ways, their completions generate two orthogonal but connected real subspaces inside a single canonical $\mathbb{C}$.

The AC dependence of $\mathbb{Q}$ is inherited from $\mathbb{Z}$ and resolves once $\mathbb{Z}$ is generated canonically. No new AC is introduced at the rational level.

4.3 The Reals: Two Completions, Two Fixed Loci

The integers generate a discrete, orientation-sensitive structure carrying two tracks. The rationals extend both tracks consistently. The real line arises from completing these tracks — but because there are two tracks, there are two completions, and they are orthogonal.

Track 1 completion: the real axis. Dedekind completion of $\mathbb{Q}$ in the additive sense collapses all holonomy information into order. A Dedekind cut (L, U) identifies every rational approach to the same boundary as the same real number. The holonomy accumulated by traversal is erased: two sequences approaching the same limit from opposite directions are identified. The result is the real axis — a copy of $\mathbb{R}$ with trivial holonomy, flat parallel transport, and maximal identification. This is the fixed-point locus of the imaginary involution $t \mapsto -t$.

Track 2 completion: the critical line. Completion of $\log(\mathbb{Q}_+) = \bigoplus_p \mathbb{Z} \cdot \log p$ in the imaginary direction fills the imaginary axis densely and completes to a copy of $\mathbb{R}$ embedded in the imaginary direction. This completion is the closure of all prime-logarithm combinations — the limit of the multiplicative prime structure as infinitely many primes contribute. The result is a copy of $\mathbb{R}$ orthogonal to the real axis, carrying the imaginary holonomy structure. Its fixed-point locus under the canonical real involution $s \mapsto 1 - s$ is the critical line $\operatorname{Re}(s) = \frac{1}{2}$.

Theorem 4.5 (Two canonical copies of $\mathbb{R}$ in the Möbius framework). *The Möbius connection generates two canonically determined copies of $\mathbb{R}$ embedded orthogonally in $\mathbb{C}$:*

- (i) **The real axis** $\{\operatorname{Im}(z) = 0\}$: *the Dedekind completion of $\mathbb{Q}$ under Track 1, the fixed-point locus of complex conjugation $z \mapsto \bar{z}$, carrying trivial holonomy.*

- (ii) **The critical line** $\{\operatorname{Re}(z) = \frac{1}{2}\}$: the completion of $\log(\mathbb{Q}_+)$ under Track 2, the fixed-point locus of the canonical involution $s \mapsto 1 - s$, carrying the imaginary holonomy structure generated by the prime logarithms.

Both copies are canonically determined by the Möbius connection without AC. Their orthogonality in $\mathbb{C}$ is forced by the independence of the two tracks, which is itself a consequence of the fundamental theorem of arithmetic.

Proof sketch. (i) is the standard Dedekind characterization with the holonomy interpretation of Remark 4.6. The fixed-locus characterization follows from $z \mapsto \bar{z}$ being the canonical orientation-reversal involution of the Möbius connection.

(ii) The set $\log(\mathbb{Q}_+)$ is dense in $\mathbb{R}$ (by linear independence of $\{\log p\}$ over $\mathbb{Q}$ and density of $\mathbb{Z}$ -linear combinations). Its completion is a copy of $\mathbb{R}$ embedded in the imaginary direction. The fixed-point locus of $s \mapsto 1 - s$ acting on $\mathbb{C}$ is $\{s : s = 1 - s\} = \{\operatorname{Re}(s) = \frac{1}{2}\}$. This is precisely where the Track 2 completion meets its own mirror image under the real involution.

The orthogonality of (i) and (ii) follows from the fact that Track 1 generates structure in the σ direction (real part) while Track 2 generates structure in the t direction (imaginary part), and these directions are orthogonal by the structure of $\mathbb{C}$. $\square$

Remark 4.6 (Trivial holonomy and the flatness of $\mathbb{R}$). *Both copies of $\mathbb{R}$ are flat in the Möbius sense: parallel transport along either copy is trivial. This is the signature of a completed structure — one that has absorbed all the holonomy information its generating mechanism produced.*

For the real axis (Track 1 completion): the holonomy $(-1)^n$ of $\mathbb{Z}$ is absorbed into the sign of the real numbers. No further holonomy remains to track. The only continuous automorphism of $\mathbb{R}$ is the identity.

For the critical line (Track 2 completion): the prime-logarithm winding is absorbed into the position along the line. The critical line is a copy of $\mathbb{R}$ — real-valued, flat, with trivial local holonomy — even though it sits at the fixed locus of the non-trivial global involution $s \mapsto 1 - s$.

This is why the Hardy Z-function — the restriction of $\zeta(s)$ to the critical line — is real-valued: the critical line is a copy of $\mathbb{R}$, and a canonical section restricted to a real subspace must be real. The phase collapse along the critical line is not a construction imposed from outside but a consequence of the Track 2 completion being a real subspace. This is developed in the companion paper [12].

Remark 4.7 (Why $\mathbb{R}$ appears twice). *The appearance of $\mathbb{R}$ at two orthogonal positions in $\mathbb{C}$ is not a duplication but a structural necessity. $\mathbb{C}$ requires two independent real directions to be a two-dimensional space over $\mathbb{R}$. The Möbius framework identifies what these two directions are and why they are not arbitrary:*

The real axis is Track 1's completion — the additive integer structure completed to a continuum.

The critical line is Track 2's completion — the multiplicative prime structure completed to a continuum.

These two completions are orthogonal because the two tracks are independent (the fundamental theorem of arithmetic makes them consistent but not identical). $\mathbb{C}$ is generated by their coexistence: the holonomy operation i is the map that takes Track 1's $\mathbb{R}$ to Track 2's $\mathbb{R}$ by a quarter-traversal, and $i^2 = -1$ because two half-traversals under the Möbius connection return $(-1)^2 \cdot (-1)^2 \dots$ no, return holonomy -1 once per full traversal. The imaginary unit is the holonomy generator relating the two real subspaces, not an algebraic extension added from outside.

4.4 The Complex Numbers: Coexistence of Two Tracks

Theorem 4.8 (Canonical generation of $\mathbb{C}$). *The complex field $\mathbb{C}$ is the structure generated by the coexistence of the two orthogonal completions of $\mathbb{Q}$ produced by the Möbius connection. It is unique without AC: the connection is unique (Theorem 3.3), the two tracks are unique (forced by the additive and multiplicative structures of $\mathbb{Z}$), and the holonomy operation relating them is unique.*

The standard description — $\mathbb{C}$ as $\mathbb{R}$ with i adjoined, satisfying $i^2 = -1$ — is correct but does not explain why $\mathbb{C}$ has the structure it does or why i exists. The Möbius framework provides the explanation.

$\mathbb{C}$ requires two real directions. The framework identifies what these directions are:

- The real axis (Track 1's $\mathbb{R}$): the Dedekind completion of $\mathbb{Q}$, generated by the additive integer structure, fixed by $z \mapsto \bar{z}$.
- The imaginary direction (Track 2's $\mathbb{R}$): the completion of $\log(\mathbb{Q}_+)$, generated by the multiplicative prime structure, whose fixed locus under $s \mapsto 1 - s$ is the critical line $\operatorname{Re}(s) = \frac{1}{2}$.

These two directions are orthogonal because the two tracks are independent (the fundamental theorem of arithmetic makes them consistent but not identical). The imaginary unit i is the holonomy operation that takes Track 1's $\mathbb{R}$ into Track 2's $\mathbb{R}$: it is the quarter-traversal of the Möbius connection, the operation that rotates the real axis into the imaginary direction. This is why $i^2 = -1$: two quarter-traversals complete a half-traversal, which under $\omega = \frac{1}{2}d\theta$ accumulates holonomy $e^{i\pi} = -1$.

The imaginary unit is not an algebraic object adjoined to $\mathbb{R}$. It is the holonomy generator of the Möbius connection, the same generator that produces the $\mathbb{Z}_2$ holonomy from which the entire number hierarchy is built. $\mathbb{C}$ is not $\mathbb{R}$ extended — it is what the Möbius connection *is*, expressed as a number system.

The Steinitz–AC result shows that two non-isomorphic algebraic closures of $\mathbb{R}$ are consistent with ZF without AC. The Möbius construction resolves this not by adding an axiom but by deriving $\mathbb{C}$ as the unique first-order limit of the two-track process: there is only one connection $\omega = \frac{1}{2}d\theta$, it generates only one pair of orthogonal tracks, and those tracks have only one consistent coexistence structure, which is $\mathbb{C}$.

Number system	Standard AC use	Type	Möbius repair
$\mathbb{Z}$	Orientation choice	Compensatory	Holonomy eigenvalue
$\mathbb{Q}$	Inherited from $\mathbb{Z}$	Upstream/hidden	Two-track pivot
$\mathbb{R}$	Ultrafilter, Vitali sets	Structural	Two fixed-point loci
$\mathbb{C}$	Algebraic closure uniqueness	Critical	Two-track coexistence

Table 1: AC dependence across the number hierarchy and its canonical repair by $\omega = \frac{1}{2}d\theta$. The Möbius connection generates not a linear chain but a two-track braid whose coexistence produces $\mathbb{C}$ as a first-order limit.

5 The Two-Track Structure and the Zeta Function

The categorical construction of the natural numbers as the initial algebra of the endofunctor $X \mapsto 1 + X$ (Lawvere's natural number object [9]) and the result that every elementary $(\infty, 1)$ -topos has

a natural number object constructed from the loop space of S^1 [10] converge on the same picture: the integers are winding numbers, and the circle comes first.

In the Möbius framework, the circle is not a generic S^1 but the S^1 equipped with $\omega = \frac{1}{2}d\theta$. The loop space ΩS^1 generates $\mathbb{Z}$ with orientation already embedded in the connection. The integers emerge from the connection already carrying both structures identified in Remark 4.2: the additive winding-number structure (Track 1) and the multiplicative prime-factorization structure (Track 2).

5.1 The Möbius Function as Holonomy Trace

The arithmetic Möbius function $\mu(n)$, defined by $\mu(n) = (-1)^k$ when n is a product of k distinct primes and $\mu(n) = 0$ otherwise [11], is the holonomy trace of the two-track structure. The sign $(-1)^k$ records the holonomy accumulated by traversing k distinct prime steps in Track 2's multiplicative structure. The squarefree condition ($\mu = 0$ for n with repeated prime factors) reflects the collapse of holonomy under repeated traversal of the same irreducible step — a step traversed twice returns $(-1)^2 = 1$, which is the identity holonomy, contributing no new orientation information.

5.2 The Zeta Function as Two-Track Section

The Riemann zeta function encodes both tracks simultaneously:

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1}. \quad (1)$$

The Dirichlet series $\sum n^{-s}$ is the Track 1 expression: it sums over all positive integers with weights n^{-s} , encoding the additive integer structure. The Euler product $\prod_p (1 - p^{-s})^{-1}$ is the Track 2 expression: it takes one factor per prime, encoding the multiplicative prime structure.

Their equality is not an analytic coincidence. It is the fundamental theorem of arithmetic expressed as an analytic identity. Every term n^{-s} in the Dirichlet series decomposes as:

$$n^{-s} = e^{-\sigma \log n} \cdot e^{-it \log n} \quad \text{where } s = \sigma + it,$$

with $\log n = \sum_k a_k \log p_k$ for the prime factorization $n = \prod_k p_k^{a_k}$. The real factor $e^{-\sigma \log n}$ is Track 1 content (integer magnitude). The complex rotation $e^{-it \log n}$ is Track 2 content (prime factorization via $\log n$ as a $\mathbb{Z}$ -linear combination of prime logarithms).

Every term in $\zeta(s)$ thus decomposes into a Track 1 component and a Track 2 component, with the fundamental theorem of arithmetic ensuring that these two components are always consistent. The function $\zeta(s)$ is the object that simultaneously encodes both tracks. It is the canonical section of the structure that $\mathbb{C}$ itself is.

5.3 The Holonomy Identity

The identity

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}$$

is the holonomy eigenvalue series of the two-track structure. The Möbius function $\mu(n) = (-1)^k$ assigns to each squarefree integer the holonomy accumulated by its prime factorization. The product $\zeta(s) \cdot \sum \mu(n)n^{-s} = 1$ is the statement that ζ and its holonomy inverse together trivialize the bundle — they span the fiber completely, with one encoding the section and the other encoding its holonomy complement.

This identity, together with the two-track structure, connects this paper’s canonical framework to the analytic theory of $\zeta(s)$ and motivates the companion paper’s treatment of the Riemann Hypothesis: the zeros of $\zeta(s)$ are the points where the two-track section vanishes, and the structure of the canonical $\mathbb{C}$ forces those vanishing points to the fixed loci of the two identification structures — the real axis (trivial zeros) and the critical line (non-trivial zeros).

6 The Reappearance of $\mathbb{C}$ Within the Tower

6.1 A Departure from Naive Expectation

The standard picture of the normed division algebra tower $\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$ treats each level as containing the previous one: $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H} \subset \mathbb{O}$. On this picture, $\mathbb{C}$ is encountered once, at level 2, and the tower proceeds upward from there. $\mathbb{C}$ is a level passed through, not a structure that recurs.

This picture is correct as a statement about the linear containment of algebras. It is misleading as a statement about the structural content of the tower. The Cayley-Dickson construction at each level does not merely extend the previous algebra: it generates, within the new level, multiple distinct copies of every lower-level algebra, each with its own orientation and parameterization relative to the level’s coordinate structure.

This reappearance is not incidental. It is a direct consequence of the Möbius connection’s generative mechanism, and it has precise implications for what the AC freedom over “which $\mathbb{C}$ ” actually amounts to — implications that are invisible in the standard presentation.

6.2 Multiple Copies of $\mathbb{C}$ in the Tower

The quaternion algebra $\mathbb{H}$ is a four-dimensional real vector space with basis $\{1, i, j, k\}$ satisfying $i^2 = j^2 = k^2 = ijk = -1$. Within $\mathbb{H}$, every two-dimensional subalgebra containing 1 and a unit imaginary element is a copy of $\mathbb{C}$. There are infinitely many such copies, parametrized by the unit sphere S^2 in the imaginary subspace of $\mathbb{H}$: for every unit imaginary quaternion $\mathbf{u}$ with $\mathbf{u}^2 = -1$, the subalgebra $\{a + b\mathbf{u} : a, b \in \mathbb{R}\}$ is isomorphic to $\mathbb{C}$.

Three of these copies are distinguished by the Cayley-Dickson structure: the planes spanned by $\{1, i\}$, $\{1, j\}$, and $\{1, k\}$. Each is a canonical copy of $\mathbb{C}$, but the three copies have different orientations within $\mathbb{H}$: the i -plane, j -plane, and k -plane are mutually perpendicular subalgebras of $\mathbb{H}$. No single one of them is preferred by the $\mathbb{H}$ -level structure; each is as canonical as the others within $\mathbb{H}$.

Proposition 6.1 ($\mathbb{H}$ contains three canonical copies of $\mathbb{C}$). *The quaternion algebra $\mathbb{H}$, as generated by the Cayley-Dickson doubling of $\mathbb{C}$, contains exactly three copies of $\mathbb{C}$ that are distinguished by the doubling structure: the i -plane, the j -plane, and the k -plane. These three copies are mutually non-isomorphic as oriented subalgebras of $\mathbb{H}$ — they share the same algebraic structure but differ in their orientation relative to the $\mathbb{H}$ -level imaginary directions. Each is a valid copy of $\mathbb{C}$, each generates a valid critical line as fixed-point locus of its conjugation involution, and none is privileged by the $\mathbb{H}$ -level structure alone.*

At the octonionic level, the structure is richer. $\mathbb{O}$ contains seven imaginary units $\{e_1, \dots, e_7\}$, each squaring to -1 , arranged according to the Fano plane. The seven $\mathbb{C}$ -copies $\{a + be_k : a, b \in \mathbb{R}\}$ and the seven $\mathbb{H}$ -copies are embedded with specific mutual orientations that reflect the Fano plane’s combinatorial structure.

Beyond $\mathbb{O}$, zero divisors appear at the sedenion level and all subsequent Cayley–Dickson algebras: the algebraic encoding capacity of the normed division algebra framework breaks down.

However, the geometric structure does not collapse — it changes character. The zero divisors organize into a constraint geometry with exact structural laws: connected components with closed-form counting formulas, functional tiers organized by the Fano plane, kernel conservation ($\dim \ker = 2 \cdot \text{degree}$, exact at every level), spectral rigidity (singular values exactly $\{0, 1, \sqrt{2}\}$), and topological protection of component boundaries. This post-NDA constraint geometry is the subject of Paper IV, which establishes the tower’s generative mechanism, and Paper V, which characterizes its functional architecture and verifies that it satisfies the axioms of an elementary topos. The Fano plane persists as the organizing principle at every post-NDA level, and the octonionic addressing system remains permanently stable through at least dimension 512 (six levels of exhaustive computation). The argument of this paper and the companion paper (Paper II) relies only on the $\mathbb{H}$ - and $\mathbb{O}$ -level structure, but the post-NDA continuation provides independent evidence for the canonicity of the connection: the same geometric object that generates the normed division algebra tower also generates the constraint geometry that organizes its aftermath.

6.3 The AC Freedom, Relocated

In the standard framework, the axiom of choice is used to select which algebraic closure of $\mathbb{R}$ to call $\mathbb{C}$. The Steinitz theorem establishes that without AC, this selection is not determined. With AC, a selection is made — but AC does not specify *which* selection, only that one exists.

The Möbius connection resolves this at the base level: it provides a canonical procedure that generates exactly one $\mathbb{C}$, without requiring a selection. The i -plane of $\mathbb{H}$ — or more precisely, the copy of $\mathbb{C}$ generated by the Möbius holonomy at the first doubling stage — is the canonical $\mathbb{C}$. There is no AC freedom in its construction.

But the AC freedom is not *eliminated*. It is *relocated*.

The copies of $\mathbb{C}$ that AC was choosing among in the standard framework correspond precisely to the multiple $\mathbb{C}$ -copies embedded in $\mathbb{H}$, $\mathbb{O}$, and the higher tower levels. Each such copy is an equally valid $\mathbb{C}$, algebraically indistinguishable from the canonical one, but embedded with a different orientation in the higher structure. The orientation freedom — which direction is “positive imaginary” — was implicitly choosing among these embedded copies, but without the tower structure to make this explicit.

Remark 6.2 (The AC freedom as tower-level multiplicity). *The standard AC freedom over “which $\mathbb{C}$ ” is not a freedom within a single level. It is the freedom to choose which tower-level copy of $\mathbb{C}$ to treat as the base-level one. Each choice produces the same $\mathbb{C}$ algebraically — the Steinitz theorem is correct that the algebraic closures are isomorphic — but they differ in how higher-level structure projects onto them.*

The Möbius connection fixes the base-level choice canonically. The other choices are visible at the $\mathbb{H}$ - and $\mathbb{O}$ -levels as the non-canonical $\mathbb{C}$ -copies with distinct orientations. These copies are not phantom alternatives: they are genuine subalgebras of the tower, each with their own holonomy structure. At post-octonionic levels, analogous structure persists in a changed form: the zero-divisor constraint geometry (Papers IV–V) organizes copies and their interactions through the Fano plane, with the octonionic addressing system remaining permanently stable. The arguments of this paper and its companion are grounded in the $\mathbb{H}$ - and $\mathbb{O}$ -level copies, but the post-NDA characterization confirms that the pattern continues with exact structural laws.

Crucially, the non-canonical $\mathbb{C}$ -copies are not independent sources of structure that stand alongside the canonical $\mathbb{C}$. They are reached from the canonical $\mathbb{C}$ by the holonomy operations that generate $\mathbb{H}$ and $\mathbb{O}$. Their critical lines are therefore not independent of the canonical critical line — they are images of the canonical critical line under those holonomy operations.

The Gamma factor in the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the structural completion that absorbs this higher-level holonomy into the canonical section. It is not a correction applied after the fact to account for tower-level complications, but a necessary component of the section's canonical form: the factor that makes ξ well-defined as a two-track section of the Möbius connection. The completed section satisfies the functional equation $\xi(s) = \xi(1-s)$ — the self-consistency condition of the canonical involution — and its zeros are exactly the non-trivial zeros of ζ . The companion paper develops this in full.

6.4 The Self-Referential Limit of Canonical Specification

The Möbius connection resolves the AC freedom at the level of *algebraic type*: it fixes which algebra appears at the base level and establishes that it is canonical. It does not resolve the freedom at the level of *metric realization*: among all geometric realizations of the canonical $\mathbb{C}$, all isomorphic as field extensions but differing in how the higher tower embeds into them, the connection does not designate one.

This is a genuine structural limitation. The base-level $\mathbb{C}$ is self-referentially related to the tower above it: the tower is built from $\mathbb{C}$, but $\mathbb{C}$'s complete metric specification requires the tower. The zeros forced by the canonical identification structure are insensitive to this limitation: their location on the critical line is fixed by algebraic type, not metric realization.

The companion paper [12] establishes that the Gamma factor is the structural completion of the canonical two-track section: $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies $\xi(s) = \xi(1-s)$, the self-consistency condition of the canonical involution. The Gamma factor absorbs the higher-level holonomy structure (the $\mathbb{H}$ - and $\mathbb{O}$ -level involutions) into the section's definition, making ξ well-defined as a global section of the Möbius connection. The self-referential limit is real — complete tower specification is not achievable from within — but it does not prevent establishing that zeros are bounded to the critical line, because the completed section's functional equation constrains zeros to the involution's fixed locus.

6.5 Structural Consequences for Zero Classification

The plane coverage argument of the companion paper [12] establishes that the canonical $\mathbb{C}$, under its two identification structures, is fully identified with the critical line through traversal: every point in the canonical $\mathbb{C}$ is reachable from the critical line by the identification group. This means the critical line is not a distinguished subset of the canonical $\mathbb{C}$ — it is the canonical $\mathbb{C}$ fully unfolded.

The reappearance of $\mathbb{C}$ in the tower must be understood in light of this. The non-canonical $\mathbb{C}$ -copies embedded in $\mathbb{H}$ and $\mathbb{O}$ are reached from the canonical $\mathbb{C}$ by holonomy operations. Their critical lines are images of the canonical critical line under those operations. A zero appearing to lie off the canonical critical line, if it originates from a non-canonical $\mathbb{C}$ -copy, is not a zero from an independent source projecting in from outside. It is a zero of the canonical structure, transformed by a holonomy operation.

The Gamma factor of the companion paper is the structural completion operator: it absorbs the higher-level holonomy — including the operations that reach the non-canonical $\mathbb{C}$ -copies — into the canonical section's definition. The completed section $\xi(s) = \xi(1-s)$ is a global section of the Möbius connection whose functional equation is the self-consistency condition of the canonical involution. The zeros of ξ are constrained to the involution's fixed locus (the critical line) by the structure of the completed section, not by a separate argument about which involutions are or are not encountered by analytic continuation.

Remark 6.3 (The bounded computation and the self-referential limit). *The two-track structure and the plane coverage argument together establish a precise constraint: zeros of $\zeta(s)$ are bounded to the critical line within whatever the Gamma factor covers. This is a constraint on the computation, not a specification of its output.*

The explicit location of zeros on the critical line — which specific values of $\text{Im}(s)$ produce simultaneous Track 1 and Track 2 cancellation — requires explicit prime computation. Knowing that zeros are restricted to the critical line does not immediately identify the zeros, just as knowing that a sorting algorithm produces sorted output does not specify which sorted list it produces for a given input. The primes are the input to the computation; the critical line is the type of the output; the specific zeros are the computed values.

The structural argument establishes that the computation process is fundamentally bounded: it cannot produce a zero outside the critical line, whatever the primes compute. This is the contribution of the geometric argument — not to determine the zeros explicitly, but to establish that the determination process is constrained to the critical line.

The companion paper [12] establishes that the Gamma factor is the structural completion that makes the canonical section well-defined: the completed $\xi(s) = \xi(1-s)$ is a global section whose functional equation constrains zeros to the involution's fixed locus. The self-referential limit — that complete tower specification is not achievable from within — does not prevent establishing the zero bound. It explains why explicit zero locations require prime computation rather than structural derivation.

7 Conclusions and Open Problems

7.1 Summary

We have established:

1. The standard constructions of $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$ carry orientation gaps: choices of direction that the algebraic constructions require but do not canonically supply. For $\mathbb{C}$ specifically, the construction $\mathbb{R}[x]/(x^2 + 1)$ produces two algebraically indistinguishable roots of $x^2 + 1 = 0$; selecting which is i is a geometric choice not determined by the algebra (Corollary 2.6, Remark 2.8).
2. The Möbius connection $\omega = \frac{1}{2}d\theta$ is the unique flat connection on S^1 whose holonomy is simultaneously discrete, irreducible, and involutive (Theorem 3.3). This uniqueness is elevated to a universal property: (S^1, ω) is the initial object in the category **OAS** of oriented arithmetic structures, which formalizes the selection (Theorem 3.9).
3. The generation pathway $\mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ is not a linear chain but a two-track braid. $\mathbb{Z}$ carries both an additive structure (winding counts, Track 1) and a multiplicative structure (prime factorizations, Track 2) from the moment of its generation. $\mathbb{Q}$ extends both tracks consistently, with the fundamental theorem of arithmetic as the unifying theorem. $\mathbb{R}$ appears twice — once as each track's completion, orthogonally embedded in $\mathbb{C}$ as the real axis (Track 1) and the critical line (Track 2). $\mathbb{C}$ is generated by the coexistence of these two orthogonal completions, with the imaginary unit i as the holonomy operation relating them (Theorems 4.1, 4.5, 4.8).
4. The Riemann zeta function is the canonical section of this two-track structure: its Dirichlet series encodes Track 1 and its Euler product encodes Track 2, with their equality being the fundamental theorem of arithmetic expressed analytically. The holonomy identity $1/\zeta(s) =$

$\sum \mu(n)n^{-s}$ is the holonomy eigenvalue series of the connection acting on prime factorizations (Section 5).

5. The canonicity of the connection is evidenced by its extension coherence: the same generative mechanism produces $\mathbb{H}$ and $\mathbb{O}$ coherently, reproducing the normed division algebra tower at precisely the known dimensions (Remark 3.12).
6. The AC freedom over “which $\mathbb{C}$ ” is not eliminated — it is relocated. The canonical $\mathbb{C}$ is the base-level copy generated by the two-track coexistence. The other choices that AC was permitting become the multiple $\mathbb{C}$ -copies embedded in $\mathbb{H}$ and $\mathbb{O}$, each with its own orientation relative to the base level (Section 6).
7. The two-track structure identifies the natural division of labor between this paper and its companion. This paper establishes the canonical $\mathbb{C}$ as the first-order limit of the Möbius connection’s processes: the two tracks, their completions, and their coexistence. The companion paper takes this completed $\mathbb{C}$ as its foundation, identifies five structural features of the canonical $\mathbb{C}$ absent from the standard algebraic construction (generated structure, identification structures, non-orientability, dense orbit structure, Gamma factor as structural completion), and shows that the combination of non-orientability and dense orbits gives a topological localization theorem: any non-trivial section compatible with the identification group vanishes only at the fixed loci. The zeros of $\zeta(s)$, as the canonical two-track section, are thereby forced to the real axis (trivial zeros) and the critical line (non-trivial zeros).

7.2 Open Problems

Problem 1 (Scope of geometric completion). Determine the full extent of geometric structure that the Möbius connection specifies in foundations of analysis beyond what is identified here — that is, map completely what the algebraic constructions leave geometrically silent and what the connection supplies.

Problem 2 (Metric content of the tower). The canonical $\mathbb{C}$ is fixed algebraically by the Möbius connection, but its metric realization — the connection lengths at each tower level, the specific orientations of the higher $\mathbb{C}$ -copies relative to the base — is not fixed by the algebraic type alone. Determine what additional conditions (consistency requirements between tower levels, physical calibration, or structural constraints on the Hopf fibration sequence) are sufficient to fix the metric content.

Problem 3 (Zero structure at higher tower levels). The zero structure of $\zeta(s)$ at the $\mathbb{C}$ level — trivial zeros at Track 1’s fixed locus, non-trivial zeros at Track 2’s fixed locus — is established by the companion paper [12]. The analogous question at the $\mathbb{H}$ level is open: what functions naturally live on the $\mathbb{H}$ -level structure, what identification structures do they carry, and what is their zero classification? The connection noted in the companion paper between trivial zeros and integers, non-trivial zeros and primes, suggests the $\mathbb{H}$ level may reveal a new class of arithmetic objects.

Problem 4 (Post-NDA constraint geometry, partially resolved). Past the octonionic level, zero divisors appear in the sedenions and all subsequent Cayley–Dickson algebras. Papers IV and V establish that these zero divisors organize into a rich constraint geometry with exact structural laws (component decomposition, functional tiers, Fano-plane organization, kernel conservation, topos structure) verified through dimension 512. The remaining open questions are: (a) whether the counting formulas established computationally extend provably to all levels via an inductive argument; (b) the precise relationship between the post-NDA constraint geometry and stable homotopy theory; and (c) the physical interpretation of the constraint architecture’s functional tiers. The

topology of the generating mechanism continues into the constraint geometry as a change of mode (from division algebra to constraint geometry), not as a termination.

Problem 5 (Langlands correspondence). Determine whether the Möbius connection provides a natural geometric framework for the L-functions appearing in the Langlands program, and whether the reappearance of $\mathbb{C}$ in the tower corresponds to the family structure of automorphic L-functions.

7.3 Companion Paper

A companion paper [12] applies this canonical foundation to the Riemann Hypothesis. Its central argument identifies five structural features of the canonical $\mathbb{C}$ that are absent from the standard algebraic construction: generated structure, identification structures, non-orientability, dense orbit structure, and the Gamma factor as structural completion. The combination of non-orientability (Feature 3) and dense orbit structure (Feature 4) yields a topological localization theorem: any non-trivial section compatible with the canonical identification group vanishes only at the fixed loci. The Gamma factor categorically exhausts all holonomy forced by analytic continuation. $\zeta(s)$, as the canonical two-track section defined by the integers, inherits the identification structure by construction and its zeros are thereby forced to the real axis (trivial zeros) and the critical line (non-trivial zeros).

The companion paper takes the two-track generation of $\mathbb{C}$ established here as its direct foundation. The geometric constitution of $\mathbb{C}$ — which this paper establishes — is precisely what the companion paper’s zero argument requires.

7.4 Methodological Note

This research was conducted in collaboration with large language model systems (Claude by Anthropic, ChatGPT by OpenAI, Gemini by Google, and Microsoft Copilot) as intellectual partners. The author is an independent researcher. All mathematical claims are the author’s responsibility.

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